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Unpacking discourses about the transition from school to university mathematics: an intensive reading

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Abstract
This paper has two main goals: first, to analyze current research related with transition to identify the foci being produced and reproduced by research through a literature review and, second, to map the connections between circulating discourses to unfold the discursive network that supports them via an intensive reading. Specifically, we are interested in how social and economic factors, educational background, and cognitive development impact the transition process and the documented potential challenges students may face. In this fashion, the questions that guide the development of the paper are as follows: What are the dominant narratives in mathematics education research about the transition from school to university mathematics? How do dominant narratives entangle particular rationalities to configure a discursive network about the transition from school to university mathematics? As a result, we identify three dominant narratives entangled, which shape a system of reason that regulates what is possible to do, act, and think.

Keywords Transition · School mathematics · University mathematics · Dominant narratives

Mathematics has been posited as a valuable resource for promoting social and economic progress (Pais, 2017) and has historically played a quintessential role in how the subject is discursively shaped into the social fabric through dominant narratives, cultural practices, and rationalities (Tröhler, 2015). The Organization for Economic Cooperation and Development (OECD) acknowledges that mathematics—as well as reading and science—is fundamental for developing societies because it is crucial in enabling students to be at the vanguard of a competitive, knowledge-based global economy (OECD, 2014). However, the abilities and competencies in mathematics developed at school may not directly align with

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the needed and required in tertiary education. The transition from school mathematics to university mathematics has become a challenge for many continuing to tertiary education, mainly because students are expected to move "from a position where concepts have an intuitive basis founded on experience to one where they are specified by formal definitions and their properties reconstructed through logical deductions" (Tall, 1992, p. 495).

This transition has established a particular focus of interest in mathematics education (see Zazkis & Holton, 2005; Di Martino et al., 2023a). Several studies address this transition as problematic, given that it requires multiple shifts—focus, epistemology, or ways of working—produced when starting university-level courses (Gueudet, 2008; Hefendehl-Hebeker, 2017; Tall, 2008). For example, Selden (2011) challenges the "rules" that govern school practices to the ones in university courses by emphasizing the "many changes in the didactic contract (Brousseau, 1997) when students move from secondary school to undergraduate study in mathematics or from undergraduate to graduate study in mathematics" (p. 391). These changes involve, for instance, the new didactical contract that privileges students' autonomy and self-regulated learning (Gueudet, 2008). Di Martino et al. (2023a) identified four main research themes through a systematic literature review of transition research published in conference proceedings, journal papers, and book chapters. These themes are (1) the mathematical gap between secondary school and university, (2) the potential of technology to facilitate the secondary-tertiary transition, (3) the factors correlated with academic success, and (4) failure in secondary-tertiary transition. Studying the transition from school to university mathematics becomes key because, building on Lerman's (2014) argument, modern society values success in mathematics and increased achievement levels. Mathematics should ground the possibilities for tertiary studies and favor the development of competencies, in Gutstein's words (2012), for reading and writing the world. Still, the relevance placed on mathematics in schools is based on ideas and arguments that position mathematics as performing a gatekeeping role.

By mapping dominant narratives, this paper problematizes the system of reason that regulates what is possible to do, say, and think about the transition from school to university mathematics. We aim to challenge dominant narratives about the transition from school to university mathematics through a critical and reflexive questioning—intensive reading (Deleuze, 1995)—of mathematics education research. The intensive reading involves focusing on recurrences of statements, as well as how these recurrences produce and reproduce a discursive network. These statements are not merely a sentence; it is not a *projection of signs*. According to Foucault (1972), each statement has an *enunciative function* independent of the subject making the statement, for example, the author of a text. A discursive statement entails rules of formation that are also domains in which it operates as an enunciative function. Then, a statement may not have the structure of a sentence or the structure of a proposition; instead, a statement has a *referent*, a *subject*, an *associated field*, and a *materiality*. Our interest is in unpacking power/knowledge relationships that make possible particular ways of writing, speaking, and thinking about the transition from school to university mathematics. Therefore, we focus on the dispersion and regularities of statements about this transition. By building on Foucault's (1972) call,

> [w]hat one must characterize and individualize is the coexistence of these dispersed and heterogeneous statements; the system that governs their division, the degree to which they depend upon one another, the way in which they interlock or exclude one another, the transformation that they undergo, and the play of their location, arrangement, and replacement. (p. 34)

We guide our analysis according to the following questions: What are the dominant narratives in mathematics education research about the transition from school to university mathematics? How do dominant narratives entangle particular rationalities to configure a discursive network about transitioning from school to university mathematics?

1 Analytical strategy

We followed two strategies to approach the questions above. First, we conducted a literature review to assemble an overview of the transition in existing research in the field, and second, we performed an intensive reading (Deleuze, 1995) to unpack dominant discourses circulating in mathematics education research about the transition from school to university mathematics. The literature review provides a snapshot of current research. It helps locate paragraphs containing enunciations about the transition from school to university mathematics in academic papers published in journals included in Web of Science and Scopus citation databases. Identifying these paragraphs enables us to find the main areas of interest in mathematics education research about the transition from school to university mathematics in order to map dominant discourses. Following an intensive reading technique, we explore how these main areas are entangled to provide a discursive network. The discursive network traces the connections between arguments and statements circulating in the selected paragraphs located in the literature review. This analysis enables us to trace recurrences and co-occurrences of arguments and statements: the dominant discourses. By identifying these dominant discourses, we are able to unfold the system of reason (Popkewitz, 2004) that regulates what is possible to do, act, and think in mathematics education research (Pais & Valero, 2012).

1.1 Systematic review to identify corpus of paragraphs on transition

The systematic review is conducted using the PRISMA method, as described by Page et al., (2021), which provides a set of procedures to ensure the replicability and transparency of the review process. First, we located the articles that would serve as our data source. We selected articles explicitly focusing on the transition from school to university mathematics from a systematic search on Web of Science and Scopus databases, given that these indexations have become an indisputable measure of research quality (see Andrade-Molina et al., 2020). Although we are not interested in questioning their quality, we are aware that papers under such indexation are more visited and cited than others. Also, scholars seem more inclined to publish their research in journals included in both indices because of their high recognition among mathematics educators and institutions, such as universities, as noted in Williams and Leatham's (2017) study. Second**,** we considered papers published within the last 12 years (2010–2021), given that it is possible to observe an increment of papers targeting the transition from school to university mathematics starting in 2008. Since 2010, this increment has been stable and constitutes an optimal cut-off point for an updated snapshot of the dominant narratives circulating in mathematics education research.

The search with the search engines for the two databases considered the combinations of keywords both in (1) English: "transition" and "mathemat*"; and "transition" and "university", and in (2) Spanish: "transición" and "matemát*"; and "transición" and "universidad." Ninety-seven articles were retrieved: 39 from Web of Science and 59 from Scopus. We removed the duplicates. Once the articles were located, we read the abstracts to discard those not focusing on the transition to university mathematics. The empirical material was

reduced to 71 articles (Table 1) (see supplementary material for the complete distribution table of papers per journal and the list of papers).

Third, both authors read the 71 articles separately to select paragraphs referencing the transition. Each paragraph is read carefully to identify its focus, specifically ideas and arguments. We only selected paragraphs containing enunciations about the transition. We triangulated the results—paragraphs identified—and made changes when necessary. As an example, after reading the materials, we selected paragraphs explicitly enunciating that students' motivation (Bengmark et al., 2017; Ufer et al., 2016) and beliefs (Bengmark et al., 2017; Geisler & Rolka, 2021; Liebendörfer & Schukajlow, 2017) are key when transitioning toward tertiary education. We divided these paragraphs into groups (numbered 1 and 6, respectively). Statements about the role played by the mathematics teacher (either in school and university teacher) within the transition were a frequent concern and constituted two additional groups (4 and 9, respectively). Lastly, we identified an interest in centering the discussion on students: specifically, their interest in mathematics (Kosiol et al., 2018; Liebendörfer & Schukajlow, 2017; Ufer et al., 2016) (5), their preparation (Frank & Thompson, 2021; Lyakhova & Neate, 2018; Nortvedt & Siqveland, 2018) (8 and 13), and their difficulties (Anastasakis et al., 2020; Frank & Thompson, 2021; Rach & Ufer, 2020) (14). We also identified groups with having a more immediate relation to each other, such as study habits (Bengmark et al., 2017), attitude toward mathematics (Bardelle & Di Martino, 2012), identity (Hernandez-Martinez et al., 2011; Wake, 2011), or self-concept (Rach & Heinze, 2017). We decided to group these into the same group, labeled "personal issues" (group numbered 17).

We are not interested in stating which focus is repeated the most or is more relevant to the field; instead, we aim to explore what mathematics education research tells about the transition from school to university mathematics. We take each group as a valid and relevant concern to the field. The paragraphs were sorted by their focus into 19 groups, which means that all 71 papers approach the transition mainly from (in no particular order): (1) students' beliefs, (2) successful transition, (3) nature of mathematics, (4) university teachers, (5) interest in mathematics, (6) students' motivation, (7) (diverse) changes, (8) students under-prepared, (9) general aspects of teachers, (10) curriculum issues, (11) community of learners, (12) gaps, (13) (effect/role) school, (14) students' difficulties, (15) new competencies/knowledge/skills, (16) students' adaptability, (17) personal issues, (18) textbooks, and (19) institutional issues.

However, some paragraphs are included in multiple groups as they explicitly engage with diverse foci. The paragraph below, for example, was included in three groups: student's beliefs (1), interest in mathematics (5), and personal issues (17):

> For academic research, we confirmed the theoretically derived hypothesis that the mathematical **belief** system—more precisely application **beliefs**—can predict the development of **interest** in the first year at university. In addition, we found that in the first academic year of teacher education, interest does not necessarily decline. For the teaching of mathematics at university, we furthermore conclude that apart from improving support for **students' autonomy and competence** (Krapp 2005; Liebendörfer and Hochmuth 2015, 2016; Schukajlow and Krug 2014b; Schukajlow et al. 2015), not only are university **students' application beliefs** important as an outcome in teacher education (Maaß 2006), but such beliefs should also be addressed to support **students' interest. (**Liebendörfer & Schukajlow, 2017, p. 363, emphasis added)

Table 1 List of papers per journal and years of publication: first five places

Source	Number of papers	Year of publications
Teaching Mathematics and Its Applications	10	2021X2–2020–2019X2–2018–2017–2016–2014–2010
ZDM — Mathematics Education	7	2021X2–2019X2–2017X2–2012
International Journal of Mathematical Education in Science and Technology	5	2020–2017–2012X2–2010
International Journal of Research in Undergraduate Mathematics Education	5	2019X3–2018X2
International Journal of Science and Mathematics Education	4	2021–2019X2–2017

1.2 Intensive reading to unpack dominant discourses on transition

As seen in the paragraph above, the focus, ideas, and arguments within a group are not isolated from other groups; instead, these are entangled with each other. In the same way that Deleuze and Guattari (1987) understand a book, the paragraphs selected are composed of "lines of articulation or segmentarity, strata, and territories; but also lines of flight, movements of deterritorialization and destratification" (p. 3). All this leads us to think that the paragraphs—their ideas and arguments—are connected to other paragraphs. We re-read the paragraphs, and, this time, we also paid attention to the connections enunciated in each paragraph, namely how focuses in paragraphs connect (*intensive reading strategy*). For Deleuze (1995), reading is as a series of ongoing experiments of creation; it is a way of thinking and reading as a concept-subjectivity-generating machine. An intensive reading from this perspective poses different questions, such as "does it work, and how does it work? How does it work for you?" (p. 8). Intensive reading is not an established set of steps that can be followed. It is a more experimental analytical move that helps see what comes through a piece of writing, as expressed by Deleuze: "If it doesn't work, if nothing comes through, you try another book. This second way of reading's intensive: something comes through or it doesn't. There's nothing to explain, nothing to understand, nothing to interpret" (p. 8). Intensive reading can be explained as placing "the text into conversation with the rest of the reader's world. [Which] introduces multiplicity [and] produces difference and invites us to examine the experiential dimension of the text[…]. The text is weighted toward the intensities and flows of the reader" (Musser, 2012, p. 137). In other words, readers engage with a text—the same text—in different ways. Reading entails a negotiation process between readers and the text, as well as writing the text, that poses a negotiation between the writer and the text. Depending on their positioning and social, cultural, and political conditions, two readers might have different interpretations of the same text given their different experiences in life and background. These can even be opposite understandings that, at the same time, might differ significantly from the writer's own understanding of the text. The narratives—through the recurrences and co-occurrences of arguments and statements—entangle specific rationalities that shape a discursive network. The discursive network is what comes through; it is what holds together the different narratives and understandings. We consider discourses as "something which produces something else[…], rather than something which exists in and of itself and which can be analysed in isolation" (Mills, 2001, p. 17). Therefore, by mapping the discursive networks, we aim to focus on

the power/knowledge relationships that make possible a particular way of writing about the transition. We are not interested in the grammatical rules of writing or the possible interpretations of the articles.

In this fashion, we are interested in exploring what comes through the articles. Using the above paragraph as an example, we highlighted these two connections: (1) *Beliefs can predict interest*; (2) *beliefs improve students' autonomy and competence*.

> For academic research, we confirmed the theoretically derived hypothesis that the mathematical **belief** system—more precisely application **beliefs—<u>can predict</u>** the development of **interest** in the first year at university. In addition, we found that in the first academic year of teacher education, **interest** does not necessarily decline. For the teaching of mathematics at university, we furthermore conclude that **<u>apart from improving</u>** support for **students' autonomy and competence** (Krapp 2005; Liebendörfer and Hochmuth 2015, 2016; Schukajlow and Krug 2014b; Schukajlow et al. 2015), **<u>not only are</u>** university **students' application beliefs** important as an outcome in teacher education (Maaß 2006), but such **beliefs should also be addressed** to support **students' interest**. (Liebendörfer & Schukajlow, 2017, p. 363, emphasis added)

In this paragraph, the underlined text highlights the two connections; beliefs connect to interest in two ways, one as a predictor of students' interests and the other as having an impact on (supporting) and improving students' interests. Locating these connections allows us to identify which groups are more frequently connected with each other, suggestive of a stronger connection. This strategy also helps us identify which groups are less likely to be connected in the selected paragraphs, suggesting a weaker connection.

It is important to clarify that intensive reading does not mean simply reading the located paragraphs again and again to make sense of them but that this way of reading involves reading the text critically and disruptively in considering that social, cultural, and political forces are entangled and operating to produce the text. In intensive reading, one leaves out the interpretation and considers the text—in our case, the articles—as "a little non–signifying machine" (Deleuze, 1995, p. 8); from this, the "process of trying to 'discover' the 'real' meaning of the text is simply an illusory practice" (Mills, 2001, p. 68), because a non-signifying machine opposes the traditional idea of a machine as a device with a specific function or purpose, the production of meanings. Thus, in this paper, the dominant narratives about the transition can be understood as a kind of assemblage of the world rather than images or representations of the world. Following Deleuze (1995), dominant narratives about the transition become a little cog in a much more complicated external machinery. Therefore, we run a more in-depth analysis of the results. We read them rhizomatically as an assemblage of particular rationalities about modern schooling and tertiary education to construct interconnected multiplicities.

> A rhizome ceaselessly establishes connections between semiotic chains, organizations of power, and circumstances relative to the arts, sciences and social struggles. A semiotic chain is like a tuber agglomerating very diverse acts, not only linguistic, but also perceptive, mimetic, gestural, and cognitive: there is no language in itself, nor are there any linguistic universals, only a throng of dialects, patois, slang, and specialized languages. There is no speaker listener, anymore than there is a homogeneous linguistic community. (Deleuze & Guattari, 1987, p. 7)

An intensive reading permits locating the recurrences (shared statements within selected paragraphs) and the *co-occurrences* (the relationship between statements). Some articles focused on the requirements for a successful transition from school to

university mathematics, either explicitly or implicitly. These recurrences intertwine with other concerns, formulations, and ideas (see Fig. 1). Frank and Thompson (2021), for example, recognized three lines of inquiry regarding students' difficulties in successfully transitioning (group 2) from school to university. One of these "highlights students' understandings of high school mathematics as impeding or supporting their successful transition to university mathematics" (p. 547). This quote states that students' beliefs (group 1), adaptability (group 16), and interest (group 5) are key to a successful transition. In the example above, *personal aspects of students* are considered requirements for a successful transition; in other articles, other recurrences are entangled to talk about the transition. Some articles focus on changes and gaps (groups 7 and 12) that schools and universities must face (see Geisler & Rolka, 2021; Frank & Thompson, 2021; Kosiol et al., 2018). Also, it is possible to find statements such as: "dynamic beliefs are beneficial for a successful transition from school to university mathematics while static beliefs seem to be disadvantageous" (Geisler & Rolka, 2021, p. 619).

In order to identify the entanglements with the large number of paragraphs, we used Gephi (https://gephi.org), a network analysis illustrator software, and generated a graph that shows the groups' entanglements through a representation that uses nodes of different sizes and lines of different width (see Fig. 1). The graph provides a visual image of the discursive network as an assemblage. More concretely, we used the algorithms *Force Atlas 2* and *Modularity* to draw the graph. *Force Atlas 2*'s layout algorithm helps visualize a graph and interpret the data based on the interaction of forces produced by nodes and lines. Gephi's algorithm "simulates a physical system in order to spatialise a network. Nodes repulse each other like charged particles, while edges [lines] attract their nodes like springs. These forces create a movement that converges to a balanced state" (Jacomy et al., 2014, p. 2). The nodes represent groups, and the node's size is determined by the width of the lines and the number of lines connected to it. The width of the lines depends on the frequency of co-occurrences between the groups—the statements that compose them. In other words, the width of the lines feeds the nodes.

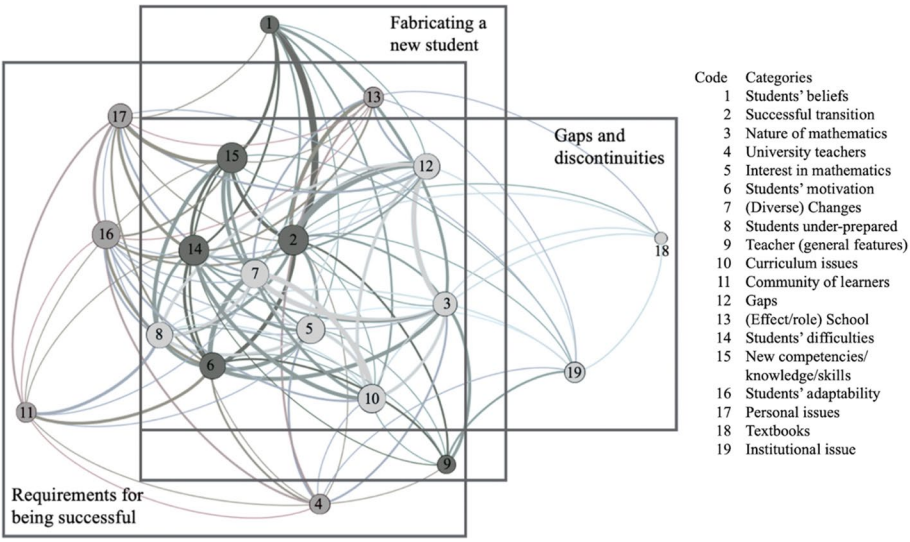

Fig. 1 Discursive network built from paragraphs located from 71 transition research papers

As noted in Fig. 1, for example, node 18 is smaller than node 3 because the lines that feed node 3 are thicker than those that feed node 18. *Modularity* helps detect communities or modules by clustering nodes that are more densely connected to each other (in light grey, grey, and dark grey in Fig. 1). Therefore, "modularity (community detection) is a measure of network structure. It was designed to measure the strength of division of a network into modules" (Ji et al., 2015, p. 1929). Modules represent interconnected groups of nodes in which connections are denser than the rest of the network (Fortunato, 2010). In looking for these denser connections, we set the modularity resolution parameter to "resolution = 0.9," "randomize," and "use weights." Higher modularity means denser connections between nodes in the same community—module—and fewer connections to nodes in different communities. Then, the nodes' size, distance, and placement trace the dominant narratives that emerge from the interactions between the groups. From this community detection algorithm, we are able to map three dominant narratives that emerged from the co-occurrences (higher-weight connections or communities in different colors in Fig. 1).

2 The discursive network: characterizing dominant narratives

The discursive network exhibits three communities with dense interconnections that shape the discourses about the transition; these are as follows: (a) *fabricating a new student* (dark gray), moving to tertiary education involves new ways of being and acting that leads to a redefinition of the self; (b) *gaps and discontinuities* (light grey), it seems that a link between what one does at school and what one does at university does not exist, due to the changes in teaching and learning. It has been acknowledged that the start of university studies implies new studying methods, tasks, and responsibilities; and (c) *requirements for being successful* (grey), mathematical knowledge has become a key element for social and personal development, given that this knowledge provides a language for science and technology, as well as a valid and valuable way of thinking.

Now that we have mapped the dominant narratives, we will illustrate them along the entanglement of the various groups, even when they belong to different dominant narratives. We will use text in bold type to denote the relevant text illustrating the entanglement.

2.1 Fabricating a new student

According to Popkewitz et al. (2017), *fabrication* is used to denote (a) a fiction quality that "direct us how to think about kinds of people in relation to changing historical circumstances" and (b) a manufacturing quality with which "narratives, stories, and theories that *manufacture*—or give a materiality to—who people are and what they do" (p. 6). We contend that this dominant narrative entails new ways of being and acting as a university student. These ways differ from the ones promoted in school. The dominant narrative fabricates different possible subjects from the relation between themself and what is desired, both in the school and university. Therefore, the move toward university studies implies facing personal and academic changes and challenges. The community mapped shows dominant narratives about students being confronted with different teaching styles and knowledge acquisition, such as, "during the transitional experience, students meet both **new knowledge and new pedagogical approaches** that they might find **challenging**" (Kouvela et al., 2017, p. 170). Other recurrences refer to the personal attributes of students. For example,

> [w]hen students come from school to university to study mathematics, they make a life-changing move. **They are expected to be more "independent" and "responsible" and embrace the feeling of "freedom" that university provides.** They move from a safety bubble created throughout their previous years of education by their school teachers and their parents to an environment that requires the **development of critical thinking, creation of consistent arguments and independent ways of thinking and working**. (Kouvela et al., 2017, pp. 165–166)

Autonomy shapes the discursive network when personal aspects of students are problematized. O'Shea and Breen (2021) highlight that in an interview with eight students, five "spoke about the necessity to be an **independent learner at university** by mentioning **working on their own or self-motivation**" (p. 36). The shared narratives about the changes students face when entering university imply that in order to become a student at university, they have to redefine themselves as students, facing demands—specific skills and knowledge—that schools have not necessarily promoted. In some cases, it is necessary to self-deconstruct to face university demands. Studying mathematics at university requires using and having new tools to meet the demands of mathematics lectures, such as more self-study than in other subjects (Lyakhova & Neate, 2021), which could become barriers or difficulties. Students' motivation, actions, subject-specific beliefs, and pre-knowledge (Bengmark et al., 2017) are part of those demands to redefine the self.

Moreover, students must expand their comfort zone to the transitional one to become more active in their learning (Kouvela et al., 2017). These changes and challenges can be understood as part of a process of alienation of students (Hernandez-Martinez, 2016). Throughout the redefinition of the university student, interest and appreciation of mathematics play a crucial role in a successful transition to university (Kosiol et al., 2018).

2.2 Gaps and discontinuities

The transition from school to university is considered a disruption instead of a continuity concerning the learning of mathematics; it is recognized that learning prerequisites related to school mathematics have a low impact on mathematics learning at university (Rach & Heinze, 2017). The mapped discursive network shows mathematics as problematic for students and instructors while transitioning from school to university (Corriveau & Bednarz, 2017; O'Shea & Breen, 2021). It is possible to find discontinuities in school and university practices. For example, the disconnection between teaching styles, such as "there is typically an **increased tempo and amount of mathematical material** to be covered (De Guzman et al., 1998), and **teaching approaches** adopted on either side of transition might not help" (Golding, 2021, p. 42). The discursive network also entangles with problems produced by high dropout rates; for example, Geisler and Rolka (2021) argue that in Germany, "up to 80% of **students quit their studies of mathematics** without graduation, 35% of them during their first year at university (Dieter & Törner, 2012) while attending courses in linear algebra and real analysis. Chen (2013) reported similar numbers for the US" (p. 599). At the same time, it is entangled with students' retention and success in higher education (see Deeken et al., 2019).

This dominant narrative is also shaped by recurrences and co-occurrences about cohort size, classroom climate at the university, new social settings, independence, and new academic requirements, becoming important features of the transitional experience (Kouvela et al., 2017). There are also interconnections pointing to the gaps produced by

mathematical knowledge and curricular structures. Despite many attempts to address and alleviate various emergent difficulties, problems persist (Kouvela et al., 2017) because school-level mathematics involves "**memorisation and computation but lacking in explanation and connections**, while university-level mathematics appeared… to be more about **theory and understanding**" (O'Shea & Breen, 2021, p. 35). In other words, the transition problem is explained as "a **misfit** between university undergraduates and the new academic learning environment" (Deeken et al., 2019, p. 25). Thus, such recurrences focus on how the ways of teaching and learning are affected. Also, study methods displace the role of students when referring to their experience while working on tasks: "the main difference [students] identified was that at second level they needed to know how to apply a formula or method, while at university level they needed to understand what they were doing and why" (O'Shea & Breen, 2021, p. 37).

2.3 Requirements for being successful

The discursive network also depicts another community, which is entangled with requirements for a successful transition. This narrative depicts unsuccessfulness in terms of high drop-out rates, especially in the early years of academic programs focusing on mathematics lectures (Kosiol et al., 2018), with which the mathematics level is one of the critical pillars for success in undergraduate science (McAlinden & Noyes, 2019). These recurrences promote initiatives that ensure students' successful transition, such as attending to subjective criteria when exploring students' drop-out: "subjective criteria of **study success** such as study **satisfaction** and student **motivation** have turned out to be early **indicators of study drop-out** beyond objective criteria such as academic achievement (Brandstätter et al. 2006; Trapmann et al. 2007)" (Kosiol et al., 2018, p. 1362). Other recurrences put their attention on interest as an essential factor in transitioning successfully, such as the ones challenging the lack of recognition: "empirical studies on the transition to university mathematics **have failed to find a relation between interest and objective criteria** of study success" (Kosiol et al., 2018, p. 1364).

It is possible to map interconnections about students needing a solid foundation of mathematical skills to make a successful transition (Lyakhova & Neate, 2018). From this, becoming a successful student at the university level is a matter of prerequisites and background. For example, the narratives that challenge how "the dual nature of mathematics as both a subject in its own right and a tool for use in other disciplines **creates difficulties in designing a suitable school curriculum** which balances these often conflicting desires" (Lyakhova & Neate, 2018, p. 167). These dominant narratives explore diverse aspects when transitioning successfully, such as expanding the scope of interventions: "**interventions in upper secondary school**, aiming at facilitating the transition to tertiary level, **must take a broad approach**. Just focusing on one aspect is not likely to be a successful strategy, especially considering the needs of different student groups" (Bengmark et al., 2017, p. 999).

Finally, some narratives place prerequisites and background at the center to question the alignment between students and the academic environment of universities. For example,

> Approaches for a **better alignment** between students and the academic environment at the universities include: (a) the **facilitation of choices of study degree programs** congruent to students' interests and abilities (see Lubinski and Benbow 2000), (b) mathematical transition c**ourses that support mathematical**

> **abilities of new STEM students** (e.g., Leviatan 2008; Griese 2017), and (c) b**etter communication and coordination** between schools and universities (recommended by Thomas et al. 2015). All such approaches for **better alignment require transparent information** on both the person and environment side. (Deeken et al., 2019, p. 27)

The recurrence of discourses not only lends density and permanence to circulating ideas but also bestows upon them value and validity. The mapped discursive network shows that some statements tend to repetitively adhere to certain boundaries to approach the transition from school to university. Foucault (1970) called this phenomenon the rarefaction of discourse. According to Foucault, discourse becomes increasingly constrained, with a select range of ideas and perspectives being repeatedly expressed and acknowledged.

3 The unpacked discourses

In this paper, we map the discursive network that holds what has been said, done, and thought about the transition. In this section, we explore the mapped discursive network to unpack the system of reason that regulates what is possible to do, say, and think about the transition from school to university mathematics. Building on Foucault and Deleuze's work, the particularities that shape our present enable us to conceive modern mathematics education as it is. In mapping the dominant narratives, we are interested in unfolding the system of reason that (re)produced circulating discourses about the transition and how these discourses shape students' subjectivities (Montecino & Andrade-Molina, Forthcoming). According to a Foucauldian approach, knowledge production is configured as a key element in governing the practices of modern society. In this light, mathematics education research produces the languages and tools that shape what we see and say within the field (Pais & Valero, 2012); it also establishes what is desirable, and, at the same time, it articulates the means for achieving such desires (Popkewitz, 2011). Therefore, the languages and specific understandings, such as the three dominant narratives, circulate through the discursive network and make possible what we think and say about the transition from school to university mathematics.

The discursive network maps the dominant narratives by detecting communities, displaying how circulating discourses become entangled when talking about the transition from school to university mathematics within the research in mathematics education. The intensive reading helped us find the stronger—denser—connections to see how dominant narratives entangle particular rationalities about the transition from school to university mathematics. By taking a closer look into the three dominant narratives—*fabricating a new student, gaps and discontinuities,* and *requirements for being successful*—it is possible to evidence the effects of power of school mathematics and university mathematics. Transitioning from school to university could be understood as a "modern-day rite of passage" by tracing new ways of being and acting (Clark & Lovric, 2009) in constantly reconfiguring the self for new events. Alternatively, as an enculturation process to a new institution, students are confronted with a new culture of how mathematics is taught and learned (Kosiol et al., 2018), which shapes students' ways of being and acting. In both cases, the discursive network produced by mathematics research plays a crucial role in the *becoming* of students, in which autonomy and self-government are operating as a route for overcoming gaps and discontinuity of students

by being "governed by and through their own interests" (Cotoi, 2011, p. 113). Although being successful is an illusion, given that the "successful subject" remains in the realm of the virtual, of immanence (Andrade-Molina, 2017), it embodies what has been historically considered as normal and desired.

After intensive reading of the nine papers published in the Special Issue entitled "Transition from School into University Mathematics: Experiences across Educational Contexts" (Di Martino et al., 2023c), we find the same three dominant narratives. For example, Lim et al. (2023) investigate changes in students' perceived norms from high school to college, and the performance on exams signals students' personal merit and status of students:

> Our findings confirmed student **struggles in college mathematics** (Alcock & Simpson, 2002; Selden, 2005) and shed light on **the students' reconstruction of social meaning of studying mathematics**, contributing to the literature (Clark & Lovric, 2009; Di Martino & Gregorio, 2019; Gueudet, 2008) regarding student perceptions of tertiary transitions in mathematics. (Lim et al., 2023, p. 73)

The differences between mathematics at school and university "are frequently recognized by students and seem to influence their **affective characteristics** towards mathematics" (Geisler et al., 2023. p. 37). The authors assert that, among others, these involve organizational as well as content-related aspects. Although we evidence other entangled narratives focusing on affective, socio-cultural, and inclusive statements (see Di Martino et al., 2023b), these do not constitute dominant narratives yet, because their interconnections—recurrences and co-occurrences—are not as strong as the three communities already mapped that shape the discursive network. However, the social and cultural aspects are beginning to be placed at the center of mathematics education concerns, for example, to explain drop-out and increase retention rates:

> Governments are also interested in **increasing retention** of mathematics students, as having a **workforce with advanced mathematics skills** is deemed to be **paramount for the development of strong economies** (Adkins & Noyes, 2016). Moreover, students with a university degree in mathematics are advantaged both in terms of **social mobility and in terms of future earnings** (for the UK context, see the report by Higher Education Statistics Agency, 2018). (Di Martino et al., 2023b, p. 80)

It appears that issues regarding students' drop-out, their difficulties in lectures, and their overall achievement increase due to unproblematized social and cultural differences.[1] For example, Deeken et al. (2019) argue that these aspects have not been considered within the transition concerns in mathematics education but configure an important part of the mismatches and difficulties in a smooth transition, to figure out the skills and competencies needed for active citizenship. FitzSimons (2019) asserts, on the one hand, that school mathematics practices of complex problem solving, critical thinking, and creativity accomplish the increasing demand for highly skilled workers. These skills become the top three skills workers need to accommodate new products, technologies, and ways of working. On the other hand, "the need (but not necessarily the desire) to study mathematics could arise as a compulsory requirement: as a condition for receiving government or other benefits, or as an entry test or hurdle requirement for employment and/or further education" (p. 43).

[1] By this, we do not mean that social and cultural differences have not been problematized at all. Instead, we mean that there are social and cultural differences that have not been problematized yet, therefore "unproblematized differences.".

The demands students face when transitioning to university, both in mathematical knowledge and individual aspects promoted by a proper background (i.e., study habits), maximize students' social and cultural differences. For example, we can think about the impact that unequal access to quality education has on the population of students coming from historically marginalized groups—ethnicity, race, economic standing, migratory status, or gender—and their performance in university-level lectures. These aspects have been acknowledged by international organizations, such as OECD and UNESCO, advising to invest in effective teaching and learning strategies at school to minimize students' gaps in mathematics. The Education 2030 Agenda, materialized in the Incheon Declaration and Framework for Action for the Implementation of Sustainable Development Goal 4 (UNESCO, 2015), states that gender equality becomes key in achieving the right to education for all:

> By 2030, eliminate gender disparities in education and ensure equal access to all levels of education and vocational training for the vulnerable, including persons with disabilities, indigenous peoples and children in vulnerable situations. (UNESCO, 2015, p. 21)

However, school mathematics focuses more on human capital and productivity than preparing students for further education in university. At school, the focus is on promoting active citizenship by developing skills such as problem solving and mathematics proficiency rather than dealing with social and cultural gaps. For example, OECD asserts that competence in mathematics.

> assists individuals in recognising the role that mathematics plays in the world and in making the well-founded judgements and decisions needed to be constructive, engaged and reflective citizens[...] Performance in mathematics described in this way encompasses more than the ability to reproduce the mathematics concepts and procedures acquired in school. (OECD, 2019, p. 104)

These desired images come from school and university, they have effects of power in shaping the discursive network and, at the same time, influence the research in mathematics education agenda and school and university curricula.

4 Conclusion

In this study, we sought to answer the following two research questions: (1) "What are the dominant narratives in mathematics education research about the transition from school to university mathematics? (2) How do dominant narratives entangle particular rationalities to configure a discursive network about transitioning from school to university mathematics?" Using an intensive reading on 71 transition research papers published between 2010 and 2021, and indexed on Web of Science and Scopus, we found three dominant narratives that entangle particular rationalities about the transition from school to university mathematics. These three dominant narratives are as follows: fabricating a new student, gaps and discontinuities, and requirements for being successful. We wonder how the fabrication of particular types of students (Montecino & Andrade-Molina, Forthcoming) prompts different nuances of productivity and achievement materialized into the desired images of successful citizens for modern society.

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Data availability The datasets analyzed during the current study (a group of 71 papers) are not publicly available due to copyright protection. However, the papers list is available in the supplementary material.

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